

Assignment 3: Question No.1 Python Task

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Create Dataset
df = pd.DataFrame({
    "Experience": [2,4,5,6,8],
    "Technical": [65,80,85,90,98],
    "Communication": [70,78,82,88,95]
})

print(df)

# Standardization
scaler = StandardScaler()

X_scaled = scaler.fit_transform(df)

print("\nStandardized Data")
print(X_scaled)

# PCA
pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

print("\nPCA Data")
print(X_pca)

print("\nExplained Variance Ratio")
print(pca.explained_variance_ratio_)

# Plot Original Data
plt.figure(figsize=(6,5))
plt.scatter(df["Technical"],
            df["Communication"],
            color="blue",
            s=80)

plt.xlabel("Technical Score")
plt.ylabel("Communication Score")
plt.title("Original Data")

plt.grid(True)
plt.show()

# Plot PCA Data
plt.figure(figsize=(6,5))
plt.scatter(X_pca[:,0],
            X_pca[:,1],
            color="red",
            s=80)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
```

```
plt.title("PCA Reduced Data")

plt.grid(True)
plt.show()
```

	Experience	Technical	Communication
0	2	65	70
1	4	80	78
2	5	85	82
3	6	90	88
4	8	98	95

Standardized Data

```
[[-1.5      -1.6850702  -1.47836825]
 [-0.5      -0.32614262 -0.53972174]
 [ 0.         0.12683324 -0.07039849]
 [ 0.5        0.5798091   0.63358639]
 [ 1.5        1.30457048  1.45490209]]
```

PCA Data

```
[[-2.69215357e+00 -1.65329079e-01]
 [-7.88852262e-01  1.58879166e-01]
 [ 3.23654719e-02  1.36854616e-01]
 [ 9.89173087e-01  2.35486571e-03]
 [ 2.45946728e+00 -1.32759569e-01]]
```

Explained Variance Ratio

```
[0.99323126 0.00592908]
```



